

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026-2027

Tutorial Sheet-6

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

- (1) Simplify the following by combining the like terms and then write whether the expression is a monomial, a binomial or a trinomial.
- (a) $7x^3 - 3x^2y + xy^2 + x^2y - y^3$
 - (b) $(x^4 + x^3y^2 + x^2y^3 + y^4) - [(-3x^3y^2 + 2x^2y^3) + (-3x^2y^3 - 5y^4)]$
 - (c) $3x^2yz^2 - 3xy^2z + x^2yz^2 + 7xy^2z$
 - (d) $50x^3 - 21x + 107 + 41x^3 - x + 1 - 93 + 71x - 31x^3$
- (2) Simplify the following expressions:
- (a) $(3x + 7)(3x - 7)$
 - (b) $(\frac{4x}{5} + \frac{y}{4})(\frac{4x}{5} + \frac{3y}{5})$
- (3) Find the value of the following expressions at $a = 1$ and $b = -2$:
- (a) $a^2 + b^2 + 3ab$
 - (b) $a^3 + a^2b + ab^2 + b^3$
- (4) If sum of first n natural numbers is found by the formula $\frac{n(n+1)}{2}$, answer the following:
- (a) Find the sum of the first 10 natural numbers.
 - (b) The sum of natural numbers from 11 to 30 (including 11 and 30).
- (5) Evaluate the following quantities using appropriate identities:
- (a) 48^2
 - (b) 497×505
 - (c) 95×105
 - (d) $181^2 - 19^2$
- (6) By using suitable identity, evaluate $x^2 + \frac{1}{x^2}$, if $x + \frac{1}{x} = 5$.
- (7) By using suitable identity, evaluate $\frac{38^2 - 22^2}{16}$.
- (8) The common factor of $17abc$, $34ab^2$, $51a^2b$ is:
- (a) $17abc$
 - (b) $17ab$
 - (c) $17ac$
 - (d) $17a^2b^2c$
- (9) The area of a rectangle is $x^2 + 12xy + 27y^2$ and its length is $(x + 9y)$. Find the breadth of the rectangle.
- (10) Factorize the following:
- (a) $21x^2y^3 + 27x^3y^2$
 - (b) $a^3 - 4a^2 + 12 - 3a$
 - (c) $4x^2 - 20x + 25$
 - (d) $\frac{y^2}{9} - 9$
- (11) Check whether $p(x)$ is a multiple of $g(x)$ or not:
- (i) $p(x) = x^3 - 5x^2 + 4x - 3$, $g(x) = x - 2$
 - (ii) $p(x) = 2x^3 - 11x^2 - 4x + 5$, $g(x) = 2x + 1$
- (12) Find the degree of the following polynomial:
- (a) $4x^4 + x^3 + 3x^5 + 5x + 7$
 - (b) $\sqrt{2}$
 - (c) $y^3(1 - y^4)$
 - (d) $\frac{8x^3}{3} - \frac{9x}{5}$
- (13) If $x + 1$ is a factor of $ax^3 + x^2 - 2x + 4a - 9$, find the value of a .

- (14) Find the zeroes of the polynomial $x^2 + \frac{1}{6}x - 2$, and verify the relation between the coefficients and the zeroes of the polynomial.
- (15) Find the zeroes of the following polynomials by factorization method and verify the relations between the zeroes and the coefficients of the polynomials:
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| (a) $4x^2 - 3x - 1$ | (b) $3x^2 + 4x - 4$ |
| (c) $5t^2 + 12t + 7$ | (d) $t^3 - 2t^2 - 15t$ |
| (e) $2x^2 + \frac{7}{2}x + \frac{3}{4}$ | (f) $4x^2 + 5\sqrt{2}x - 3$ |
| (g) $2s^2 - (1 + 2\sqrt{2})s + \sqrt{2}$ | (h) $v^2 + 4\sqrt{3}v - 15$ |
- (16) Given that the zeroes of the cubic polynomial $x^3 - 6x^2 + 3x + 10$ are of the form $a, a+b, a+2b$ for some real numbers a and b , find the values of a and b as well as the zeroes of the given polynomial.
- (17) Find k so that $x^2 + 2x + k$ is a factor of $2x^4 + x^3 - 14x^2 + 5x + 6$. Also find all the zeroes of the two polynomials.
- (18) Given that $x - \sqrt{5}$ is a factor of the cubic polynomial $x^3 - 3\sqrt{5}x^2 + 13x - 3\sqrt{5}$, find all the zeroes of the polynomial.
- (19) For which values of a and b , are the zeroes of $q(x) = x^3 + 2x^2 + a$ also the zeroes of the polynomial $p(x) = x^5 - x^4 - 4x^3 + 3x^2 + 3x + b$? Which zeroes of $p(x)$ are not the zeroes of $q(x)$?
